

Effective AI Prompting: Your Essential Skill

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- Welcome to this Spoken Tutorial on Effective Prompting for AI.



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Learning Objectives



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- Recall the three core components of an effective AI prompt.



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- Recall the three core components of an effective AI prompt.
- Apply advanced techniques like Chain-of-Thought and structured output.



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Learning Objectives

- Recall the three core components of an effective AI prompt.
- Apply advanced techniques like Chain-of-Thought and structured output.
- Create clear and specific prompts for both text and image generation.



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Prerequisites



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- Basic computer literacy.



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- Access to a Large Language Model (LLM) tool.



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- Examples: Google Gemini, ChatGPT, or Claude.



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AI as a Smart Intern: The Power of Prompting



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AI as a Smart Intern: The Power of Prompting

- AI models are like smart, fast interns.



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- They need clear instructions to do good work.



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- Prompting is giving these instructions.



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AI as a Smart Intern: The Power of Prompting

- AI models are like smart, fast interns.
- They need clear instructions to do good work.
- Prompting is giving these instructions.
- A good prompt leads to high-quality results.



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The 3 C's: Clarity



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The 3 C's: Clarity

- Good prompts almost always include:
Clarity, Context, and Constraints.



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- Clarity: Be specific! AI does not know what you are thinking.



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- Weak: "Write about electric cars."



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The 3 C's: Clarity

- Good prompts almost always include: Clarity, Context, and Constraints.
- Clarity: Be specific! AI does not know what you are thinking.
- Weak: "Write about electric cars."
- Good: "Write a short blog post explaining the top 3 benefits of owning an electric car for city residents."



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The 3 C's: Context and Constraints + Persona



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The 3 C's: Context and Constraints + Persona

- Context: Give background. Tell AI why and for whom.



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The 3 C's: Context and Constraints + Persona

- Context: Give background. Tell AI why and for whom.
- Better Prompt (with Context): "Write for a sustainable lifestyle blog. Explain top 3 benefits of electric cars for city residents."



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- Context: Give background. Tell AI why and for whom.
- Better Prompt (with Context): "Write for a sustainable lifestyle blog. Explain top 3 benefits of electric cars for city residents."
- Constraints: Set rules. Define length, format, and style.



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Giving Examples: Zero, One, and Few-Shot



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- Zero-Shot: You give zero examples. (Most common).



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- Example: "Summarize text like this: Text: 'Sun is a star.' Output: 'Primary celestial body: Sun'."



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Giving Examples: Zero, One, and Few-Shot

- Zero-Shot: You give zero examples. (Most common).
- One-Shot: You give one example to show a pattern.
- Example: "Summarize text like this: Text: 'Sun is a star.' Output: 'Primary celestial body: Sun'."
- Few-Shot: You give a few examples for



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Advanced Techniques: Chain-of-Thought (CoT)



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- Chain-of-Thought (CoT): Ask AI to "Think Step-by-Step".



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Advanced Techniques: Chain-of-Thought (CoT)

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- Useful for logic, math, reasoning problems.



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- Reduces errors by showing AI's logic.



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Advanced Techniques: Chain-of-Thought (CoT)

- Chain-of-Thought (CoT): Ask AI to "Think Step-by-Step".
- Useful for logic, math, reasoning problems.
- Reduces errors by showing AI's logic.
- Prompt: "I have 5 boxes of apples, 12 in each. I eat 3, give away 1 full box. How many left? Think step-by-step."



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Advanced Techniques: Structuring Output & Iterative Prompting



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- Structuring Output: Tell AI the exact format you need (Markdown table, HTML, JSON, list).



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- Structuring Output: Tell AI the exact format you need (Markdown table, HTML, JSON, list).
- Prompt: "List pros and cons of remote work as a Markdown table with 'Pros' and 'Cons' columns."



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Advanced Techniques: Structuring Output & Iterative Prompting

- Structuring Output: Tell AI the exact format you need (Markdown table, HTML, JSON, list).
- Prompt: "List pros and cons of remote work as a Markdown table with 'Pros' and 'Cons' columns."
- Iterative Prompting: It is a conversation!
Refine prompts based on AI's replies.



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Practical Lab: Text Generation Challenge



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Practical Lab: Text Generation Challenge

- Open Google Gemini, ChatGPT, or Claude.



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Practical Lab: Text Generation Challenge

- Open Google Gemini, ChatGPT, or Claude.
- Try this prompt: "Act as a fitness coach. Create a simple 3-day workout plan for a beginner who wants to exercise at home with no equipment. The goal is a full-body workout. Present it in a table format with columns for 'Day', 'Exercise', and 'Duration'."



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Practical Lab: Image Prompting Challenge



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Practical Lab: Image Prompting Challenge

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- Think like an artist: describe visual details.



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Practical Lab: Image Prompting Challenge

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- Try this: "A close-up, photorealistic image of a cat wearing a tiny detective hat and magnifying glass, sitting in a library. The lighting is warm and cinematic."



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Practical Lab: Image Prompting Challenge

- Image tools like Midjourney or DALL-E 3 use descriptive prompts.
- Think like an artist: describe visual details.
- Try this: "A close-up, photorealistic image of a cat wearing a tiny detective hat and magnifying glass, sitting in a library. The lighting is warm and cinematic."
- Use Midjourney or DALL-E 3 if available.



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Summary



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- Advanced techniques: Chain-of-Thought, structured output, iterative prompting



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Summary

- AI needs specific instructions (prompts).
- Good prompts have Clarity, Context, Constraints (3 C's).
- Use Persona and examples (Zero, One, Few-Shot).
- Advanced techniques: Chain-of-Thought, structured output, iterative prompting.
- Practice is key for effective prompting.



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Assignment



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- Use an AI tool (Gemini/ChatGPT/Claude).



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Assignment

- Use an AI tool (Gemini/ChatGPT/Claude).
- Prompt: "Act as a friendly tour guide for Mumbai. Create a 1-day itinerary for a family visiting Mumbai for the first time. They want to see historical places and try local street food. Include time estimates. Present this as a bulleted list with timings."



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- Observe the output and refine your prompt if needed.



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Thank You!



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- Thank you for joining this tutorial!



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